

An Overview of Animal Diversity

KEY CONCEPTS

- 32.1** Animals are multicellular, heterotrophic eukaryotes with tissues that develop from embryonic layers
- 32.2** The history of animals spans more than half a billion years
- 32.3** Animals can be characterized by “body plans”
- 32.4** Views of animal phylogeny continue to be shaped by new molecular and morphological data

▲ **Figure 32.1** What adaptations make a chameleon a fearsome predator?

A Kingdom of Consumers

Although slow-moving on its feet, the chameleon in **Figure 32.1** can wield its long, sticky tongue with blinding speed to capture its unsuspecting prey. Many species of chameleons can also change their color and thereby blend into their surroundings—making them hard to detect, both by their prey and by the animals that would eat them.

The chameleon is just one example of an animal that is an efficient consumer of other organisms. Other predatory animals overwhelm their prey using their strength, speed, or toxins, while still others capture the unwary by building concealed traps such as webs. Likewise, herbivorous animals can strip the plants they eat bare of leaves or seeds, while parasitic animals weaken their hosts by consuming their tissues or body fluids. These and other animals are effective eating machines in part because they have specialized muscle and nerve cells that enable them to detect, capture, and eat other organisms—including those that can flee from attack. Animals are also very good at processing the food they have eaten; most animals do this using an efficient digestive system that has a mouth at one end and an anus at the other.

In this chapter, we embark on a tour of the animal kingdom that will continue in the next two chapters. Here we will consider the characteristics that all animals share, as well as the evolutionary history of this kingdom of consumers.

Animals are multicellular, heterotrophic eukaryotes with tissues that develop from embryonic layers

Listing features shared by all animals is challenging, as there are exceptions to nearly every criterion we might select. When taken together, however, several characteristics of animals sufficiently describe the group for our discussion.

Nutritional Mode

Animals differ from both plants and fungi in their mode of nutrition. Plants are autotrophic eukaryotes capable of generating organic molecules through photosynthesis. Fungi are heterotrophs that grow on or near their food and that feed by absorption (often after they have released enzymes that digest the food outside their bodies). Unlike plants, animals cannot construct all of their own organic molecules, and so, in most cases, they ingest them—either by eating other living organisms or by eating nonliving organic material. But unlike fungi, most animals feed by ingesting their food and then using enzymes to digest it within their bodies.

Cell Structure and Specialization

Animals are eukaryotes, and like plants and most fungi, animals are multicellular. In contrast to plants and fungi, however, animals lack the structural support of cell walls. Instead, proteins external to the cell membrane provide structural support to animal cells and connect them to one another (see Figure 6.28). The most abundant of these proteins is collagen, which is not found in plants or fungi.

The cells of most animals are organized into **tissues**, groups of similar cells that act as a functional unit. For example, muscle tissue and nervous tissue are responsible for moving the body and conducting nerve impulses, respectively. The ability to move and conduct nerve impulses underlies many of the adaptations that differentiate animals from plants and fungi (which lack muscle and nerve cells). For this reason, muscle and nerve cells are central to the animal lifestyle.

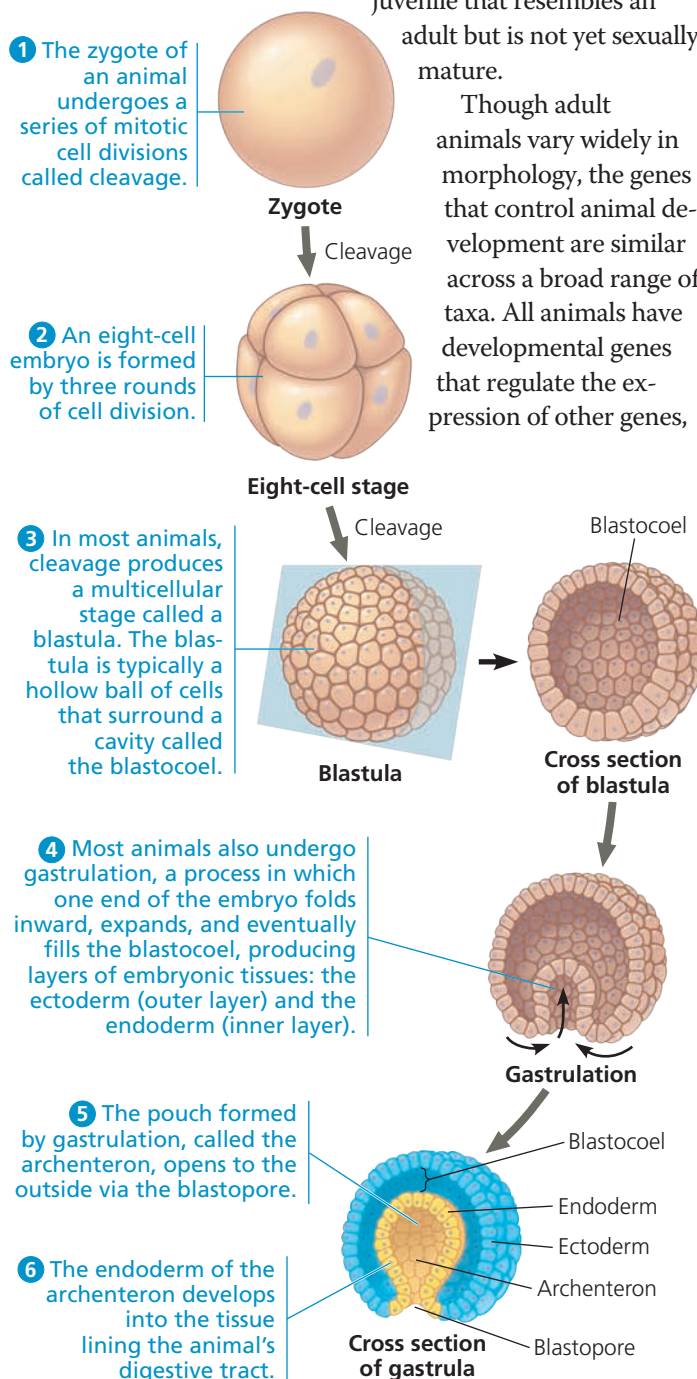
Reproduction and Development

Most animals reproduce sexually, and the diploid stage usually dominates the life cycle. In the haploid stage, sperm and egg cells are produced directly by meiotic division, unlike what occurs in plants and fungi (see Figure 13.6). In most animal species, a small, flagellated sperm fertilizes a larger, nonmotile egg, forming a diploid zygote. The zygote then undergoes **cleavage**, a succession of mitotic cell divisions without cell growth between the divisions. During the development of most animals, cleavage leads to the formation of a multicellular stage called a **blastula**, which in many animals takes the form of a hollow

ball (**Figure 32.2**). Following the blastula stage is the process of **gastrulation**, during which the layers of embryonic tissues that will develop into adult body parts are produced. The resulting developmental stage is called a **gastrula**.

Although some animals, including humans, develop directly into adults, the life cycles of most animals include at least one larval stage. A **larva** is a sexually immature form of an animal that is morphologically distinct from the adult, usually eats different food, and may even have a different habitat than the adult, as in the case of the aquatic larva of a mosquito or dragonfly. Animal larvae eventually undergo **metamorphosis**, a developmental transformation that turns the animal into a juvenile that resembles an adult but is not yet sexually mature.

Though adult animals vary widely in morphology, the genes that control animal development are similar across a broad range of taxa. All animals have developmental genes that regulate the expression of other genes,



▲ **Figure 32.2** Early embryonic development in animals.

and many of these regulatory genes contain sets of DNA sequences called *homeoboxes* (see Chapter 21). In particular, most animals share a unique homeobox-containing family of genes, known as *Hox* genes. *Hox* genes play important roles in the development of animal embryos, controlling the expression of many other genes that influence morphology.

Sponges, which are among the simplest extant (living) animals, lack *Hox* genes. However, they have other homeobox genes that influence their shape, such as those that regulate the formation of water channels in the body wall, a key feature of sponge morphology (see Figure 33.4). In the ancestors of more complex animals, the *Hox* gene family arose via the duplication of earlier homeobox genes. Over time, the *Hox* gene family underwent a series of duplications, yielding a versatile “toolkit” for regulating development. In most animals, *Hox* genes regulate the formation of the anterior-posterior (front-to-back) axis, as well as other aspects of development. Similar sets of conserved genes govern the development of both flies and humans, despite their obvious differences and hundreds of millions of years of divergent evolution.

CONCEPT 32.2

The history of animals spans more than half a billion years

To date, biologists have identified 1.3 million extant species of animals, and estimates of the actual number run far higher. This vast diversity encompasses a spectacular range of morphological variation, from corals to cockroaches to crocodiles. Various studies suggest that this great diversity originated during the last billion years. For example, researchers have unearthed 710-million-year-old sediments containing the fossilized remains of steroids that today are primarily produced by a particular group of sponges. Hence, these fossil steroids suggest that animals had arisen by 710 million years ago.

DNA analyses generally agree with this fossil biochemical evidence; for example, one recent molecular clock study estimated that sponges originated about 700 million years ago. These findings are also consistent with molecular analyses suggesting that the common ancestor of all extant animal species lived about 770 million years ago. What was this common ancestor like, and how did animals arise from their single-celled ancestors?

Steps in the Origin of Multicellular Animals

One way to gather information about the origin of animals is to identify protist groups that are closely related to animals. As shown in **Figure 32.3**, a combination of morphological

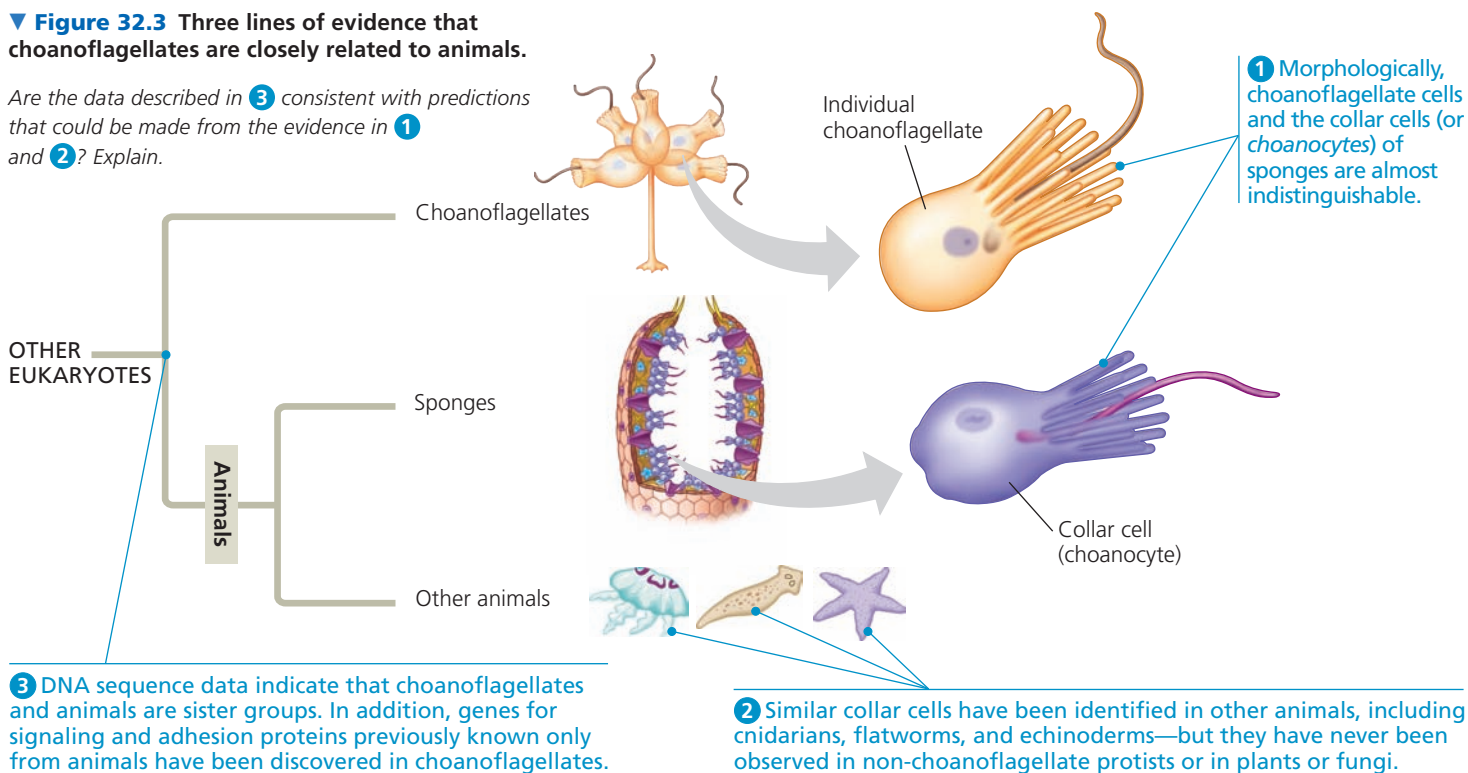
CONCEPT CHECK 32.1

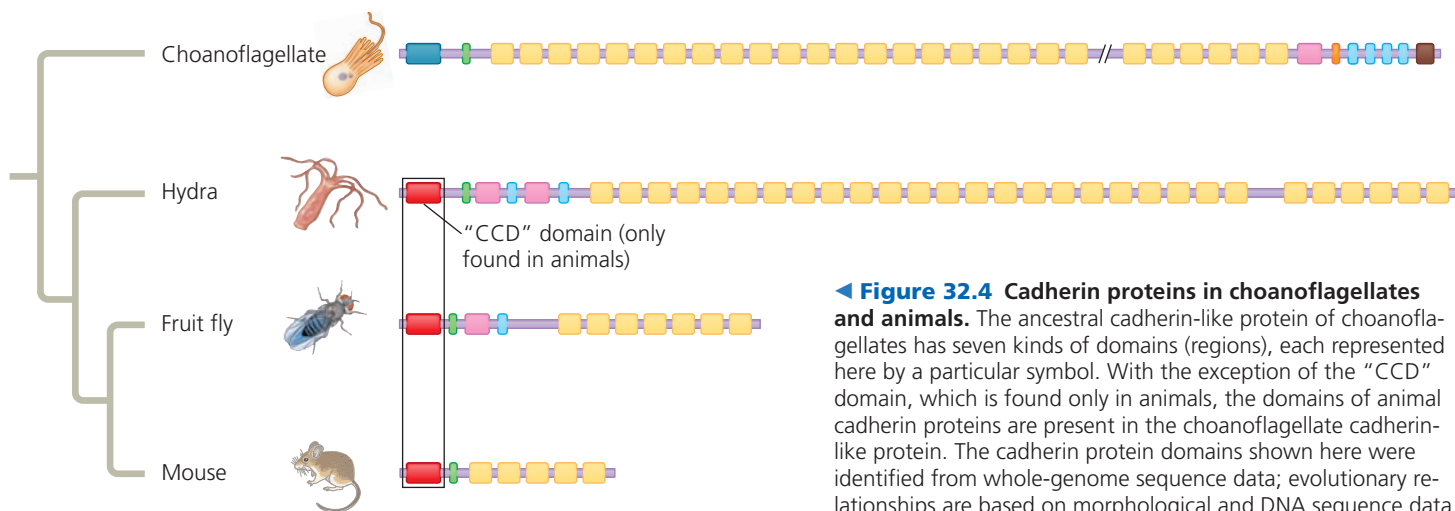
1. Summarize the main stages of animal development. What family of control genes plays a major role?
2. **WHAT IF?** What animal characteristics would be needed by an imaginary plant that could chase, capture, and digest its prey—yet could also extract nutrients from soil and conduct photosynthesis?

For suggested answers, see Appendix A.

▼ Figure 32.3 Three lines of evidence that choanoflagellates are closely related to animals.

Are the data described in **3** consistent with predictions that could be made from the evidence in **1** and **2**? Explain.





and molecular evidence points to choanoflagellates as the closest living relatives of animals. Based on such evidence, researchers have hypothesized that the common ancestor of choanoflagellates and living animals may have been a suspension feeder similar to present-day choanoflagellates.

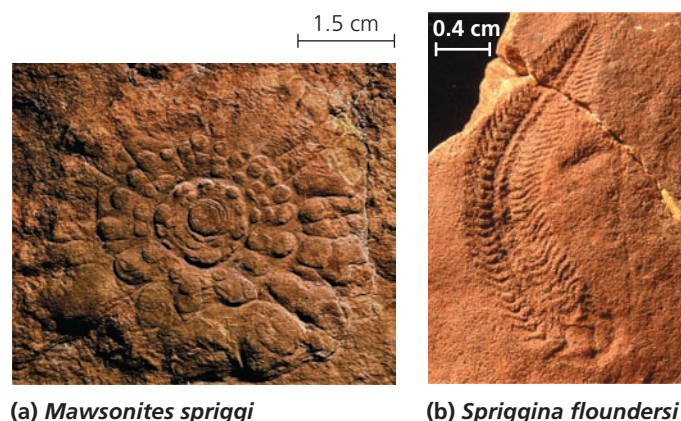
Scientists exploring *how* animals may have arisen from their single-celled ancestors have noted that the origin of multicellularity requires the evolution of new ways for cells to adhere (attach) and signal (communicate) to each other. In an effort to learn more about such mechanisms, Dr. Nicole King (featured in the interview before Chapter 26) and colleagues compared the genome of the unicellular choanoflagellate *Monosiga brevicollis* with those of representative animals. This analysis uncovered 78 protein domains in *M. brevicollis* that were otherwise only known to occur in animals. (A *domain* is a key structural or functional region of a protein.) For example, *M. brevicollis* has genes that encode domains of certain proteins (known as cadherins) that play key roles in how animal cells attach to one another, as well as genes that encode protein domains that animals (and only animals) use in cell-signaling pathways.

Let’s take a closer look at the cadherin attachment proteins we just mentioned. DNA sequence analyses show that animal cadherin proteins are composed primarily of domains that are also found in a cadherin-like protein of choanoflagellates (Figure 32.4). However, animal cadherin proteins also contain a highly conserved region not found in the choanoflagellate protein (the “CCD” domain labeled in Figure 32.4). These data suggest that the cadherin attachment protein originated by the rearrangement of protein domains found in choanoflagellates plus the incorporation of a novel domain, the conserved CCD region. Overall, comparisons of choanoflagellate and animal genomes suggest that key steps in the transition to multicellularity in animals involved new ways of using proteins or parts of proteins that were encoded by genes found in choanoflagellates.

Next, we’ll survey the fossil evidence for how animals evolved from their distant common ancestor over four geologic eras (see Table 25.1 to review the geologic time scale).

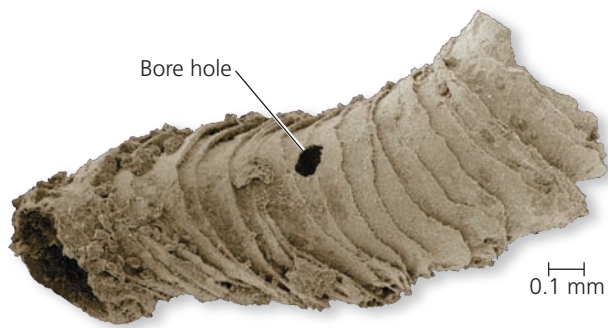
Neoproterozoic Era (1 Billion–542 Million Years Ago)

Although data from fossil steroids and molecular clocks indicate an earlier origin, the first generally accepted macroscopic fossils of animals date from about 560 million years ago. These fossils are members of an early group of soft-bodied multicellular eukaryotes, known collectively as the **Ediacaran biota**. The name comes from the Ediacara Hills of Australia, where fossils of these organisms were first discovered (Figure 32.5). Similar fossils have since been found on other continents. Among the oldest Ediacaran fossils that resemble animals, some are thought to be molluscs (snails and their relatives), while others may be related to sponges and cnidarians (sea anemones and their relatives). Still others have proved



(a) *Mawsonites spriggi* (b) *Spriggina floundersi*

▲ **Figure 32.5 Ediacaran fossil animals.** Fossils dating to about 560 million years ago include those resembling animals (a) with simple, radial forms and (b) with many body segments.



▲ **Figure 32.6 Early evidence of predation.** This 550-million-year-old fossil of the animal *Cloudina* shows evidence of having been attacked by a predator that bored through its shell.

difficult to classify, as they do not seem to be closely related to any living animal or algal groups. In addition to these macroscopic fossils, Neoproterozoic rocks have also yielded what may be microscopic fossils of early animal embryos. Although these microfossils appear to exhibit the basic structural organization of present-day animal embryos, debate continues about whether these fossils are indeed of animals.

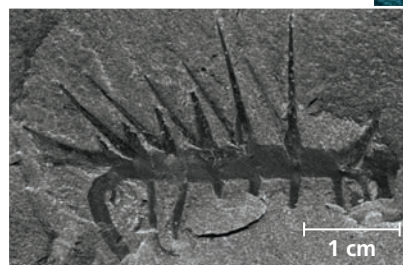
The fossil record from the Ediacaran period (635–542 million years ago) also provides early evidence of predation. Consider *Cloudina*, a small animal whose body was protected by a shell resembling a series of nested cones (Figure 32.6). Some *Cloudina* fossils show signs of attack: round “bore holes” that resemble those formed today by predators that drill through the shells of their prey to gain access to the soft-bodied organisms lying within. Like *Cloudina*, some other small Ediacaran animals had shells or other defensive structures that may have been selected for by predators. Overall, the fossil evidence indicates that the Ediacaran was a time of increasing animal diversity—a trend that continued in the Paleozoic.

Paleozoic Era (542–251 Million Years Ago)

Another wave of animal diversification occurred 535–525 million years ago, during the Cambrian period of the Paleozoic era—a phenomenon referred to as the **Cambrian explosion** (see Chapter 25). In strata formed before the Cambrian explosion, only a few animal phyla have been observed. But in strata that are 535–525 million years old, paleontologists have found the oldest fossils of about half of all extant animal phyla, including the first arthropods, chordates, and echinoderms. Many of these fossils, which include the first large animals

with hard, mineralized skeletons, look very different from most living animals (Figure 32.7). Even so, paleontologists have established that these Cambrian fossils are members of extant animal phyla, or at least are close relatives. In particular, most of the fossils from the Cambrian explosion are of **bilaterians**, an enormous clade whose members (unlike sponges and cnidarians) have a two-sided or bilaterally symmetric form and a complete digestive tract, an efficient digestive system that has a mouth at one end and an anus at the other. As we’ll discuss later in the chapter, bilaterians include molluscs, arthropods, chordates, and most other living animal phyla.

As the diversity of animal phyla increased during the Cambrian, the diversity of Ediacaran life-forms declined. What caused these trends? Fossil evidence suggests that during the Cambrian period, predators acquired novel adaptations, such as forms of locomotion that helped them catch prey, while prey species acquired new defenses, such as protective shells. As new predator-prey relationships emerged, natural selection may have led to the decline of the soft-bodied Ediacaran species and the rise of various bilaterian phyla. Another hypothesis focuses on an increase in atmospheric oxygen that preceded the Cambrian explosion. More plentiful oxygen would have enabled animals with higher metabolic rates and larger body sizes to thrive, while potentially harming other species. A third hypothesis proposes that genetic changes affecting development, such as the origin of *Hox* genes and the addition of new microRNAs (small RNAs involved in gene regulation), facilitated the evolution of new



Hallucigenia fossil (530 mya)



▲ **Figure 32.7 A Cambrian seascape.** This artist’s reconstruction depicts a diverse array of organisms found in fossils from the Burgess Shale site in British Columbia, Canada. The animals include *Pikaia* (eel-like chordate at top left), *Marella* (small arthropod swimming at left), *Anomalocaris* (large animal with grasping limbs and a circular mouth), and *Hallucigenia* (animals with toothpick-like spikes on the seafloor and in inset).

SCIENTIFIC SKILLS EXERCISE

Calculating and Interpreting Correlation Coefficients

Is Animal Complexity Correlated with miRNA Diversity?

Animal phyla vary greatly in morphology, from simple sponges that lack tissues and symmetry to complex vertebrates. Members of different animal phyla have similar developmental genes, but the number of miRNAs varies considerably. In this exercise, you will explore whether miRNA diversity is correlated to morphological complexity.

How the Study Was Done

In the analysis, miRNA diversity is represented by the average number of miRNAs in a phylum ($\bar{x}$), while morphological complexity is represented by the average number of cell types ($\bar{y}$). The researchers examined the relationship between these two variables by calculating the correlation coefficient (r). The correlation coefficient indicates the extent and direction of a linear relationship between two variables (x and y) and ranges in value between -1 and 1 . When $r < 0$, y and x are negatively correlated, meaning that values of y become smaller as values of x become larger. When $r > 0$, y and x are positively correlated (y becomes larger as x becomes larger). When $r = 0$, the variables are not correlated.

The formula for the correlation coefficient r is:

$$r = \frac{\frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})}{s_x s_y}$$

In this formula, n is the number of observations, x_i is the value of the i^{th} observation of variable x , and y_i is the value of the i^{th} observation of variable y . $\bar{x}$ and $\bar{y}$ are the means of variables x and y , and s_x and s_y are the standard deviations of variables x and y . The “ Σ ” symbol indicates that the n values of the product $(x_i - \bar{x})(y_i - \bar{y})$ are to be added together.

Interpret the Data

- First, practice reading the data table. For the eighth observation ($i = 8$), what are x_i and y_i ? For which phylum are these data?
- Next, we'll calculate the mean and standard deviation for each variable. (a) The **mean** ($\bar{x}$) is the sum of the data values divided by n , the number of observations: $\bar{x} = \frac{\sum x_i}{n}$. Calculate the mean number of

Data from the Study

Animal Phylum	i	No. of miRNAs (x_i)	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$	No. of Cell Types (y_i)	$(y_i - \bar{y})$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
Porifera	1	5.8			25			
Platyhelminthes	2	35			30			
Cnidaria	3	2.5			34			
Nematoda	4	26			38			
Echinodermata	5	38.6			45			
Cephalochordata	6	33			68			
Arthropoda	7	59.1			73			
Urochordata	8	25			77			
Mollusca	9	50.8			83			
Annelida	10	58			94			
Vertebrata	11	147.5			172.5			
		$\bar{x} =$ $s_x =$		$\Sigma =$	$\bar{y} =$ $s_y =$		$\Sigma =$	$\Sigma =$

miRNAs ($\bar{x}$) and the mean number of cell types ($\bar{y}$) and enter them in the data table (for $\bar{y}$, replace each x in the formula with a y). (b) Next, calculate $(x_i - \bar{x})$ and $(y_i - \bar{y})$ for each observation, recording your results in the appropriate column. Square each of those results to complete the $(x_i - \bar{x})^2$ and $(y_i - \bar{y})^2$ columns; sum the results for those columns. (c) The **standard deviation**, s_x , which describes the variation found in the data, is calculated using the following formula:

$$s_x = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

(d) Calculate s_x and s_y by substituting the results in (b) into the formula for the standard deviation.

- Next, calculate the correlation coefficient r for the variables x and y . (a) First, use the results in 3(b) to complete the $(x_i - \bar{x})(y_i - \bar{y})$ column; sum the results in that column. (b) Now use the values for s_x and s_y from 3(c) along with the results from 4(a) in the formula for r .
- Do these data indicate that miRNA diversity and animal complexity are negatively correlated, positively correlated, or uncorrelated? Explain.
- What does your analysis suggest about the role of miRNA diversity in the evolution of animal complexity?

 A version of this Scientific Skills Exercise can be assigned in MasteringBiology.

Data from Bradley Deline, University of West Georgia, and Kevin Peterson, Dartmouth College, 2013.

body forms. In the **Scientific Skills Exercise**, you can investigate whether there is a correlation between microRNAs (miRNAs; see Figure 18.14) and body complexity in various animal phyla. These various hypotheses are not mutually exclusive; predator-prey relationships, atmospheric changes, and changes in development may each have played a role.

The Cambrian period was followed by the Ordovician, Silurian, and Devonian periods, when animal diversity continued to increase, although punctuated by episodes of mass extinction (see Figure 25.17). Vertebrates (fishes) emerged as the top predators of the marine food web. By 450 million years ago, groups that diversified during the Cambrian period began to make an impact on land. Arthropods were the first animals to adapt to terrestrial habitats, as indicated

by fragments of arthropod remains and by well-preserved fossils from several continents of millipedes, centipedes, and spiders. Another clue is seen in fossilized fern galls—enlarged cavities that fern plants form in response to stimulation by resident insects, which then use the galls for protection. Fossils indicate that fern galls date back at least 302 million years, suggesting that insects and plants were influencing each other's evolution by that time.

Vertebrates colonized land around 365 million years ago and diversified into numerous terrestrial groups. Two of these survive today: the amphibians (such as frogs and salamanders) and the amniotes (reptiles, including birds, and mammals). We will explore these groups, known collectively as the tetrapods, in more detail in Chapter 34.

Mesozoic Era (251–65.5 Million Years Ago)

The animal phyla that had evolved during the Paleozoic now began to spread into new habitats. In the oceans, the first coral reefs formed, providing other marine animals with new places to live. Some reptiles returned to the water, leaving plesiosaurs (see Figure 25.5) and other large aquatic predators as their descendants. On land, descent with modification in some tetrapods led to the origin of wings and other flight equipment in pterosaurs and birds. Large and small dinosaurs emerged, both as predators and herbivores. At the same time, the first mammals—tiny nocturnal insect-eaters—appeared on the scene. In addition, as you read in Chapter 30, flowering plants (angiosperms) and insects both underwent dramatic diversifications during the late Mesozoic.

Cenozoic Era (65.5 Million Years Ago to the Present)

Mass extinctions of both terrestrial and marine animals ushered in a new era, the Cenozoic. Among the groups of species that disappeared were the large, nonflying dinosaurs and the marine reptiles. The fossil record of the early Cenozoic documents the rise of large mammalian herbivores and predators as mammals began to exploit the vacated ecological niches. The global climate gradually cooled throughout the Cenozoic, triggering significant shifts in many animal lineages. Among primates, for example, some species in Africa adapted to the open woodlands and savannas that replaced many of the former dense forests. The ancestors of our own species were among those grassland apes.

CONCEPT CHECK 32.2

1. Put the following milestones in animal evolution in order from oldest to most recent: (a) origin of mammals, (b) earliest evidence of terrestrial arthropods, (c) Ediacaran fauna, (d) extinction of large, nonflying dinosaurs.
2. **WHAT IF?** Suppose the most recent common ancestor of extant fungi and animals lived 1 billion years ago. If the first fungi lived 990 million years ago, would extant animals also have been alive at that time? Explain.
3. **MAKE CONNECTIONS** Evaluate whether the origin of cell-to-cell attachment proteins in animals illustrates descent with modification. (See Concept 22.2.)

For suggested answers, see Appendix A.

CONCEPT 32.3

Animals can be characterized by “body plans”

Animal species vary tremendously in morphology, but their great diversity in form can be described by a relatively small number of major “body plans.” A **body plan** is a particular set of morphological and developmental traits,

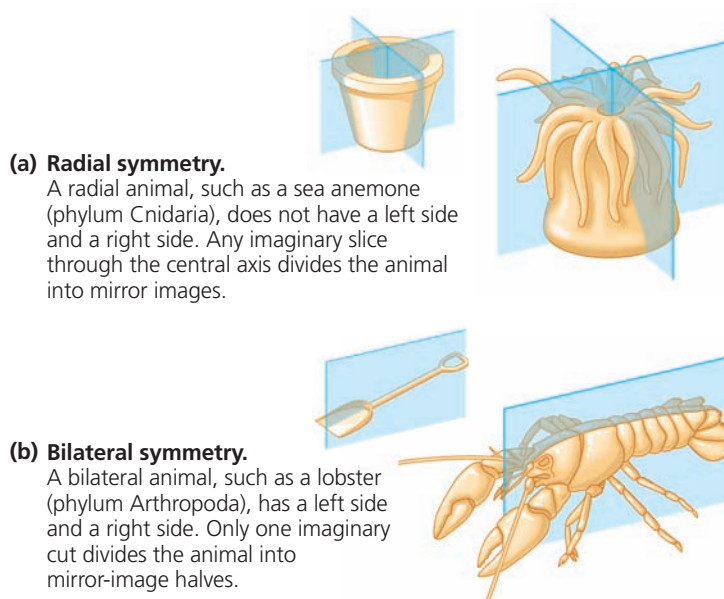
integrated into a functional whole—the living animal. The term *plan* here does not imply that animal forms are the result of conscious planning or invention. But body plans do provide a succinct way to compare and contrast key animal features. They also are of interest in the study of *evo-devo*, the interface between evolution and development.

Like all features of organisms, animal body plans have evolved over time. In some cases, including key stages in gastrulation, novel body plans emerged early in the history of animal life and have not changed since. As we’ll discuss, however, other aspects of animal body plans have changed multiple times over the course of evolution. As we explore the major features of animal body plans, bear in mind that similar body forms may have evolved independently in different lineages. In addition, body features can be lost over the course of evolution, causing some closely related species to look very different from one another.

Symmetry

A basic feature of animal bodies is their type of symmetry—or absence of symmetry. (Many sponges, for example, lack symmetry altogether.) Some animals exhibit **radial symmetry**, the type of symmetry found in a flowerpot (Figure 32.8a). Sea anemones, for example, have a top side (where the mouth is located) and a bottom side. But they have no front and back ends and no left and right sides.

The two-sided symmetry of a shovel is an example of **bilateral symmetry** (Figure 32.8b). A bilateral animal has two axes of orientation: front to back and top to bottom. Such animals have a **dorsal** (top) side and a **ventral** (bottom) side, a left side and a right side, and an **anterior** (front) end



▲ **Figure 32.8** Body symmetry. The flowerpot and shovel are included to help you remember the radial-bilateral distinction.

and a **posterior** (back) end. Many animals with a bilaterally symmetrical body plan (such as arthropods and mammals) have sensory equipment concentrated at their anterior end, including a central nervous system (“brain”) in the head.

The symmetry of an animal generally fits its lifestyle. Many radial animals are sessile (living attached to a substrate) or planktonic (drifting or weakly swimming, such as jellies, commonly called jellyfishes). Their symmetry equips them to meet the environment equally well from all sides. In contrast, bilateral animals typically move actively from place to place. Most bilateral animals have a central nervous system that enables them to coordinate the complex movements involved in crawling, burrowing, flying, or swimming. Fossil evidence indicates that these two fundamentally different kinds of symmetry have existed for at least 550 million years.

Tissues

Animal body plans also vary with regard to tissue organization. Recall that tissues are collections of specialized cells that act as a functional unit; in animals, true tissues are isolated from other tissues by membranous layers. Sponges and a few other groups lack true tissues. In all other animals, the embryo becomes layered during gastrulation. As development progresses, these layers, called *germ layers*, form the various tissues and organs of the body. **Ectoderm**, the germ layer covering the surface of the embryo, gives rise to the outer covering of the animal and, in some phyla, to the central nervous system. **Endoderm**, the innermost germ layer, lines the pouch that forms during gastrulation (the archenteron) and gives rise to the lining of the digestive tract (or cavity) and organs such as the liver and lungs of vertebrates.

Cnidarians and a few other animal groups that have only these two germ layers are said to be **diploblastic**. All bilaterally symmetrical animals have a third germ layer, called the **mesoderm**, which fills much of the space between the ectoderm and endoderm. Thus, animals with bilateral symmetry are also said to be **triploblastic** (having three germ layers). In triploblasts, the mesoderm forms the muscles and most other organs between the digestive tract and the outer covering of the animal. Triploblasts include a broad range of animals, from flatworms to arthropods to vertebrates. (Although some diploblasts actually do have a third germ layer, it is not nearly as well developed as the mesoderm of animals considered to be triploblastic.)

Body Cavities

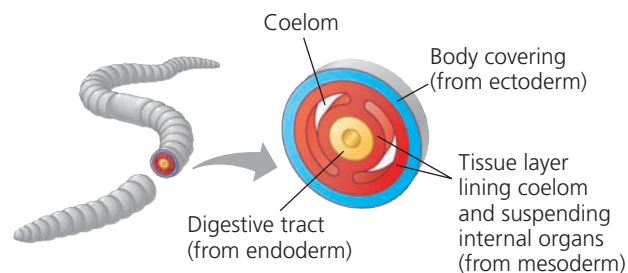
Most triploblastic animals have a **body cavity**, a fluid- or air-filled space located between the digestive tract and the outer body wall. This body cavity is also called a **coelom** (from the Greek *kóilos*, hollow). A so-called “true” coelom forms from tissue derived from mesoderm. The inner and outer layers of tissue that surround the cavity connect and

form structures that suspend the internal organs. Animals with a true coelom are known as **coelomates** (Figure 32.9a).

Some triploblastic animals have a body cavity that is formed from mesoderm and endoderm (Figure 32.9b). Such a cavity is called a “pseudocoelom” (from the Greek *pseudo*, false), and the animals that have one are called **pseudocoelomates**. Despite its name, however, a pseudocoelom is not false; it is a fully functional body cavity. Finally, some triploblastic animals lack a body cavity altogether (Figure 32.9c). They are known collectively as **acoelomates** (from the Greek *a-*, without).

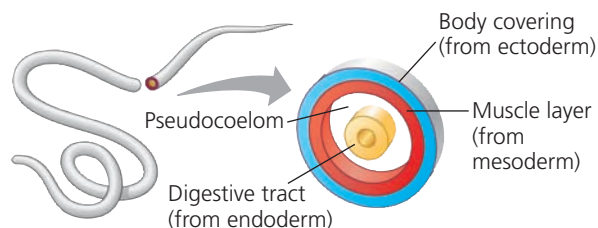
▼ **Figure 32.9** Body cavities of triploblastic animals. The organ systems develop from the three embryonic germ layers.

(a) Coelomate



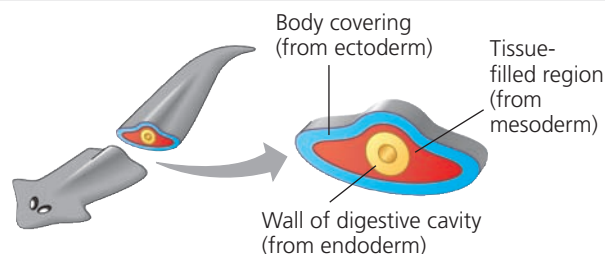
Coelomates, such as earthworms, have a true coelom, a body cavity completely lined by tissue derived from mesoderm.

(b) Pseudocoelomate



Pseudocoelomates, such as roundworms, have a body cavity lined by tissue derived from mesoderm and by tissue derived from endoderm.

(c) Acoelomate



Acoelomates, such as planarians, lack a body cavity between the digestive cavity and outer body wall.

Key

■ Ectoderm ■ Mesoderm ■ Endoderm

A body cavity has many functions. Its fluid cushions the suspended organs, helping to prevent internal injury. In soft-bodied coelomates, such as earthworms, the coelom contains noncompressible fluid that acts like a skeleton against which muscles can work. The cavity also enables the internal organs to grow and move independently of the outer body wall. If it were not for your coelom, for example, every beat of your heart or ripple of your intestine would warp your body's surface.

Terms such as *coelomates* and *pseudocoelomates* refer to organisms that have a similar body plan and hence belong to the same *grade* (a group whose members share key biological features). However, phylogenetic studies show that true coeloms and pseudocoeloms have been independently gained or lost multiple times in the course of animal evolution. As shown by this example, a grade is not necessarily equivalent to a *clade* (a group that includes an ancestral species and all of its descendants). Thus, while terms such as coelomate or pseudocoelomate can be helpful in describing an organism's features, these terms must be interpreted with caution when seeking to understand evolutionary history.

Protostome and Deuterostome Development

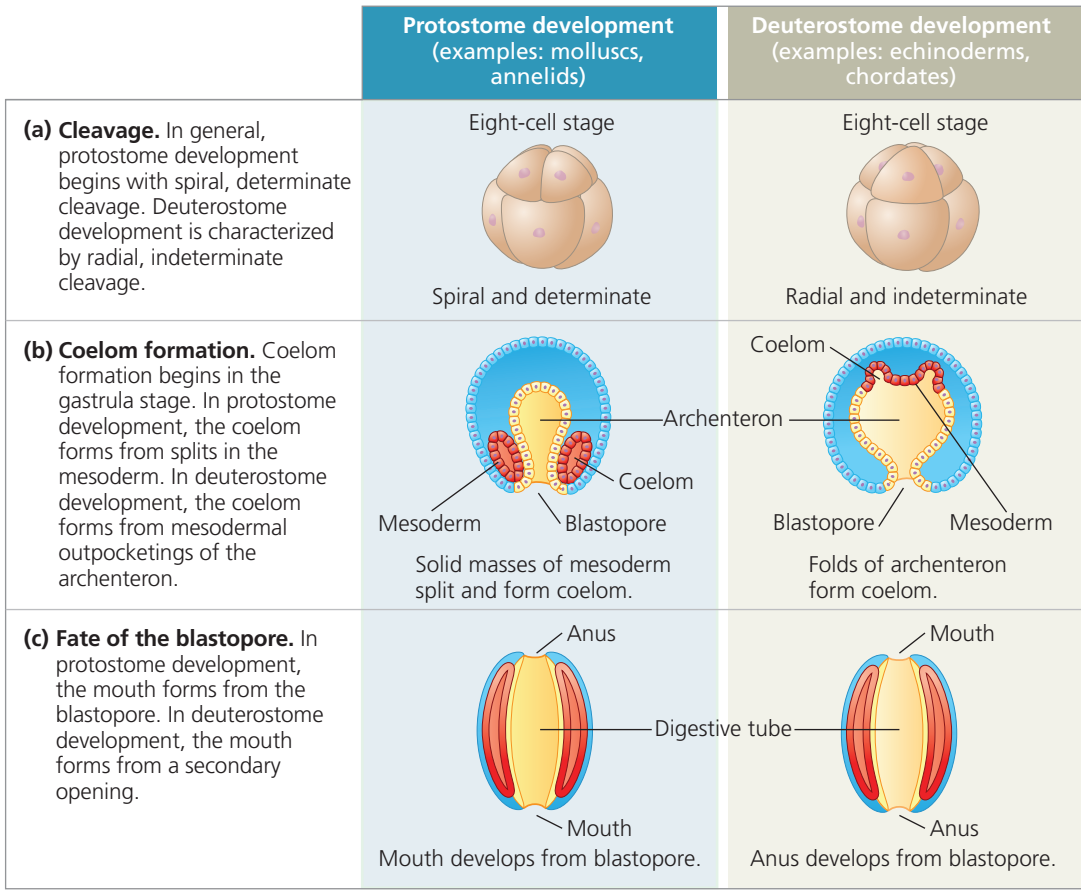
Based on certain aspects of early development, many animals can be described as having one of two developmental

modes: **protostome development** or **deuterostome development**. These modes can generally be distinguished by differences in cleavage, coelom formation, and fate of the blastopore.

Cleavage

Many animals with protostome development undergo **spiral cleavage**, in which the planes of cell division are diagonal to the vertical axis of the embryo; as seen in the eight-cell stage of the embryo, smaller cells are centered over the grooves between larger, underlying cells (**Figure 32.10a**, left). Furthermore, the so-called **determinate cleavage** of some animals with protostome development rigidly casts (“determines”) the developmental fate of each embryonic cell very early. A cell isolated from a snail at the four-cell stage, for example, cannot develop into a whole animal. Instead, after repeated divisions, such a cell will form an inviable embryo that lacks many parts.

In contrast to the spiral cleavage pattern, deuterostome development is predominantly characterized by **radial cleavage**. The cleavage planes are either parallel or perpendicular to the vertical axis of the embryo; as seen at the eight-cell stage, the tiers of cells are aligned, one directly above the other (see Figure 32.10a, right). Most animals with deuterostome development also have **indeterminate cleavage**, meaning that each cell produced by early



◀ **Figure 32.10** A comparison of protostome and deuterostome development. These are useful general distinctions, though there are many variations and exceptions to these patterns.

MAKE CONNECTIONS Re-view Figure 20.21. As an early embryo, which would more likely have stem cells capable of giving rise to cells of any type: an animal with protostome development or one with deuterostome development? Explain.

cleavage divisions retains the capacity to develop into a complete embryo. For example, if the cells of a sea urchin embryo are separated at the four-cell stage, each can form a complete larva. Similarly, it is the indeterminate cleavage of the human zygote that makes identical twins possible.

Coelom Formation

During gastrulation, an embryo's developing digestive tube initially forms as a blind pouch, the **archenteron**, which becomes the gut (Figure 32.10b). As the archenteron forms in protostome development, initially solid masses of mesoderm split and form the coelom. In contrast, in deuterostome development, the mesoderm buds from the wall of the archenteron, and its cavity becomes the coelom.

Fate of the Blastopore

Protostome and deuterostome development often differ in the fate of the **blastopore**, the indentation that during gastrulation leads to the formation of the archenteron (Figure 32.10c). After the archenteron develops, in most animals a second opening forms at the opposite end of the gastrula. In many species, the blastopore and this second opening become the two openings of the digestive tube: the mouth and the anus. In protostome development, the mouth generally develops from the first opening, the blastopore, and it is for this characteristic that the term *protostome* derives (from the Greek *protos*, first, and *stoma*, mouth). In deuterostome development (from the Greek *deuteros*, second), the mouth is derived from the secondary opening, and the blastopore usually forms the anus.

CONCEPT CHECK 32.3

1. Distinguish the terms *grade* and *clade*.
2. Compare three aspects of the early development of a snail (a mollusc) and a human (a chordate).
3. **WHAT IF?** Evaluate this claim: Ignoring the details of their specific anatomy, worms, humans, and most other triploblasts have a shape analogous to that of a doughnut.

For suggested answers, see Appendix A.

CONCEPT 32.4

Views of animal phylogeny continue to be shaped by new molecular and morphological data

As animals with diverse body plans radiated during the early Cambrian, some lineages arose, thrived for a period of time, and then became extinct, leaving no descendants. However, by 500 million years ago, most animal phyla with members

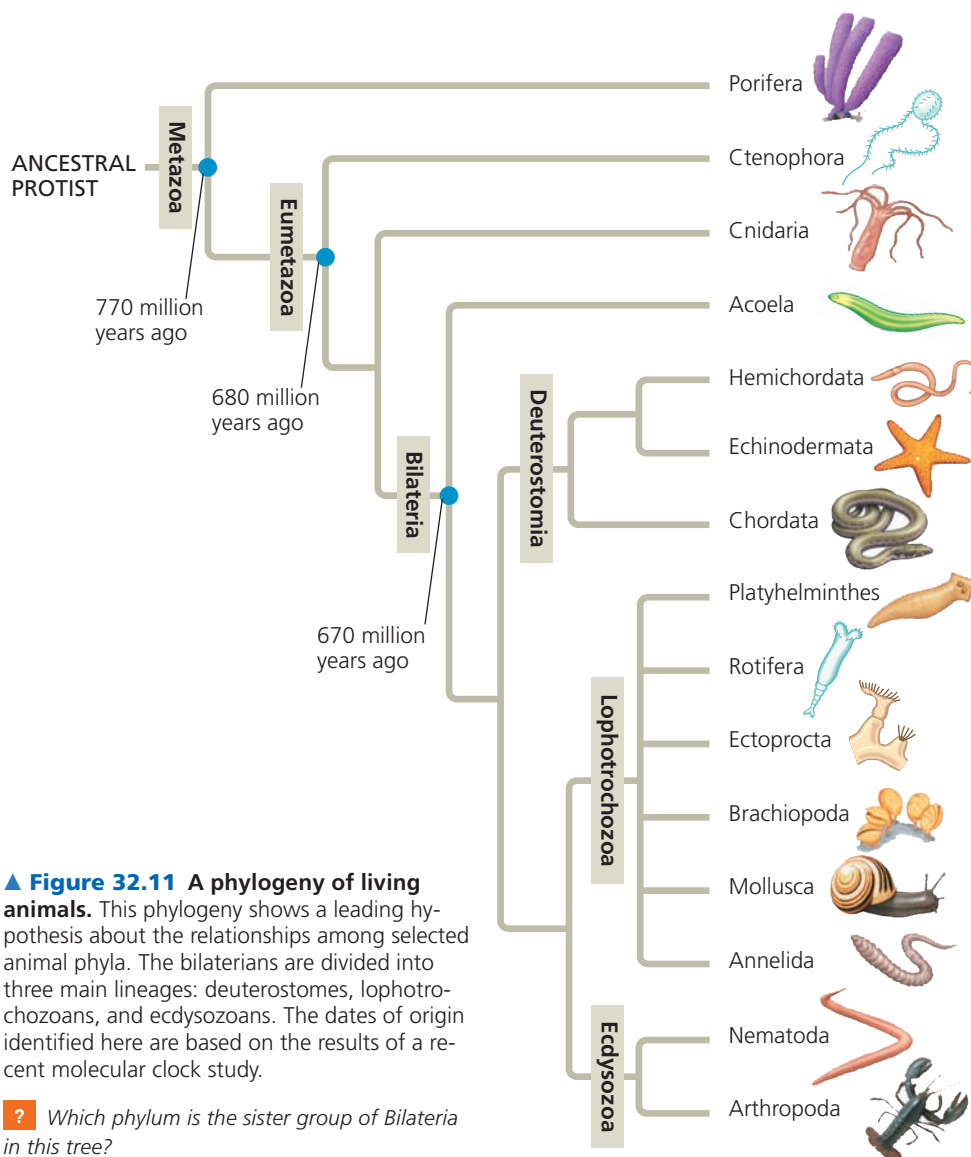
alive today were established. Next, we'll examine relationships among these taxa along with some remaining questions that are currently being addressed using genomic data.

The Diversification of Animals

Zoologists currently recognize about three dozen phyla of extant animals, 15 of which are shown in Figure 32.11. Researchers infer evolutionary relationships among these phyla by analyzing whole genomes, as well as morphological traits, ribosomal RNA (rRNA) genes, *Hox* genes, protein-coding nuclear genes, and mitochondrial genes. Notice how the following points are reflected in Figure 32.11.

1. **All animals share a common ancestor.** Current evidence indicates that animals are monophyletic, forming a clade called Metazoa. All extant and extinct animal lineages have descended from a common ancestor.
2. **Sponges are basal animals.** Among the extant taxa, sponges (phylum Porifera) branch from the base of the animal tree. Recent morphological and molecular analyses indicate that sponges are monophyletic, as shown here.
3. **Eumetazoa is a clade of animals with true tissues.** All animals except for sponges and a few others belong to a clade of **eumetazoans** ("true animals"). True tissues evolved in the common ancestor of living eumetazoans. Basal eumetazoans, which include the phyla Ctenophora (comb jellies) and Cnidaria, are diploblastic and generally have radial symmetry.
4. **Most animal phyla belong to the clade Bilateria.** Bilateral symmetry and the presence of three prominent germ layers are shared derived characters that help define the clade Bilateria. This clade contains the majority of animal phyla, and its members are known as *bilaterians*. The Cambrian explosion was primarily a rapid diversification of bilaterians.
5. **There are three major clades of bilaterian animals.** Bilaterians have diversified into three main lineages, Deuterostomia, Lophotrochozoa, and Ecdysozoa. With one exception, the phyla in these clades consist entirely of **invertebrates**, animals that lack a backbone; Chordata is the only phylum that includes **vertebrates**, animals with a backbone.

As seen in Figure 32.11, hemichordates (acorn worms), echinoderms (sea stars and relatives), and chordates are members of the bilaterian clade **Deuterostomia**; thus, the term *deuterostome* refers not only to a mode of animal development, but also to the members of this clade. (The dual meaning of this term can be confusing since some organisms with a deuterostome developmental pattern are *not* members of clade Deuterostomia.) Hemichordates share some characteristics with chordates, such as gill slits and a dorsal nerve cord; echinoderms lack these characteristics. These shared traits may have been present in the common

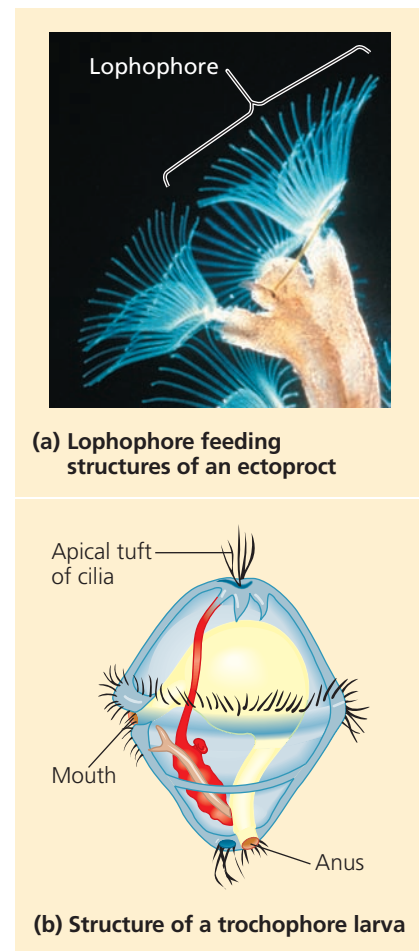


▲ **Figure 32.11 A phylogeny of living animals.** This phylogeny shows a leading hypothesis about the relationships among selected animal phyla. The bilaterians are divided into three main lineages: deuterostomes, lophotrochozoans, and ecdysozoans. The dates of origin identified here are based on the results of a recent molecular clock study.

? Which phylum is the sister group of Bilateria in this tree?

ancestor of the deuterostome clade (and lost in the echinoderm lineage). As mentioned above, phylum Chordata, the only phylum with vertebrate members, also includes invertebrates.

Bilaterians also diversified in two major clades that are composed entirely of invertebrates: the *ecdysozoans* and the *lophotrochozoans*. The clade name **Ecdysozoa** refers to a characteristic shared by nematodes, arthropods, and some of the other ecdysozoan phyla that are not included in our survey. These animals secrete external skeletons (exoskeletons); the stiff covering of a cricket and the flexible cuticle of a nematode are examples. As the animal grows, it molts, squirming out of its old exoskeleton and secreting a larger one. The process of shedding the old exoskeleton is called *ecdysis*. Though named for this characteristic, the clade was proposed mainly on the basis of molecular data that support the common ancestry of its members. Furthermore, some taxa excluded from this clade by their molecular data, such as certain species of leeches, do in fact molt.



▲ **Figure 32.12 Morphological characteristics of lophotrochozoans.**

The name **Lophotrochozoa** refers to two different features observed in some animals belonging to this clade. Some lophotrochozoans, such as ectoprocts, develop a unique structure called a **lophophore** (from the Greek *lophos*, crest, and *pherein*, to carry), a crown of ciliated tentacles that function in feeding (**Figure 32.12a**). Individuals in other phyla, including molluscs and annelids, go through a distinctive developmental stage called the **trochophore larva** (**Figure 32.12b**)—hence the name lophotrochozoan.

Future Directions in Animal Systematics

While many scientists think that current evidence supports the evolutionary relationships shown in Figure 32.11, aspects of this phylogeny continue to be debated. Although it can be frustrating that the phylogenies in textbooks cannot be memorized as set-in-stone truths, the uncertainty inherent in these diagrams is a healthy reminder that science

is an ongoing, dynamic process of inquiry. We'll conclude with three questions that are the focus of ongoing research.

1. **Are sponges monophyletic?** Traditionally, sponges were placed in a single phylum, Porifera. This view began to change in the 1990s, when molecular studies indicated that sponges were paraphyletic; as a result, sponges were placed into several different phyla that branched near the base of the animal tree. Since 2009, however, several morphological and molecular studies have concluded that sponges are a monophyletic group after all, as traditionally thought and as shown in Figure 32.11. Researchers are currently sequencing the entire genomes of various sponges to investigate whether sponges are indeed monophyletic.
2. **Are ctenophores basal metazoans?** Many researchers have concluded that sponges are basal metazoans (see Figure 32.11). However, several recent studies have placed the comb jellies (phylum Ctenophora) at the base of the animal tree. Data that are consistent with placing sponges at the base of the animal tree include fossil steroid evidence, molecular clock analyses, the morphological similarity of sponge collar cells to the cells of choanoflagellates (see Figure 32.3), and the fact that sponges are one of the few animal groups that lack true tissues (as might be expected for basal animals). Ctenophores, on the other hand, have true tissues and their cells do not resemble the cells of choanoflagellates. At present, the idea that ctenophores are basal metazoans remains an intriguing but controversial hypothesis.

3. **Are acoelomate flatworms basal bilaterians?** A series of recent molecular papers have indicated that acoelomate flatworms (phylum Acoela) are basal bilaterians, as shown in Figure 32.11. A different conclusion was supported by a 2011 analysis, which placed acoelomates within Deuterostomia. Researchers are currently sequencing the genomes of several acoelomates and species from closely related groups to provide a more definitive test of the hypothesis that acoelomate flatworms are basal bilaterians. If further evidence supports this hypothesis, this would suggest that the bilaterians may have descended from a common ancestor that resembled living acoelomate flatworms—that is, from an ancestor that had a simple nervous system, a saclike gut with a single opening (the “mouth”), and no excretory system.

CONCEPT CHECK 32.4

1. Describe the evidence that cnidarians share a more recent common ancestor with other animals than with sponges.
2. **WHAT IF?** Suppose ctenophores are basal metazoans and sponges are the sister group of all remaining animals. Under this hypothesis, redraw Figure 32.11 and discuss whether animals with true tissues would form a clade.
3. **MAKE CONNECTIONS** Based on the phylogeny in Figure 32.11 and the information in Figure 25.11, evaluate this statement: “The Cambrian explosion actually consists of three explosions, not one.”

For suggested answers, see Appendix A.

32 Chapter Review

SUMMARY OF KEY CONCEPTS

CONCEPT 32.1

Animals are multicellular, heterotrophic eukaryotes with tissues that develop from embryonic layers (pp. 668–669)

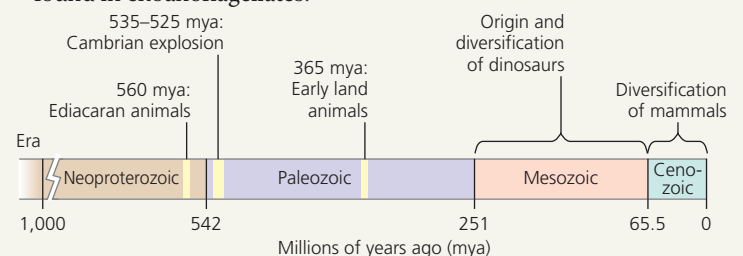
- Animals are heterotrophs that ingest their food.
- Animals are multicellular eukaryotes. Their cells are supported and connected to one another by collagen and other structural proteins located outside the cell membrane. Nervous tissue and muscle tissue are key animal features.
- In most animals, **gastrulation** follows the formation of the **blastula** and leads to the formation of embryonic tissue layers. Most animals have *Hox* genes that regulate the development of body form. Although *Hox* genes have been highly conserved over the course of evolution, they can produce a wide diversity of animal morphology.

? Describe key ways that animals differ from plants and fungi.

CONCEPT 32.2

The history of animals spans more than half a billion years (pp. 669–673)

- Fossil biochemical evidence and molecular clock analyses indicate that animals arose over 700 million years ago.
- Genomic analyses suggest that key steps in the origin of animals involved new ways of using proteins that were encoded by genes found in choanoflagellates.



? What caused the Cambrian explosion? Describe current hypotheses.

CONCEPT 32.3

Animals can be characterized by “body plans” (pp. 673–676)

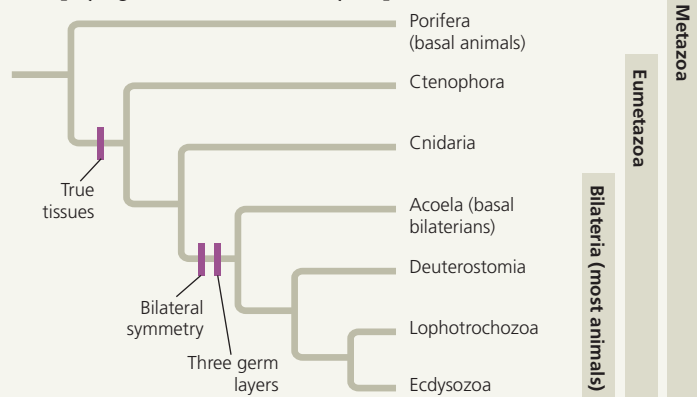
- Animals may lack symmetry or may have radial or bilateral symmetry. Bilaterally symmetrical animals have dorsal and ventral sides, as well as anterior and posterior ends.
- Eumetazoan embryos may be **diploblastic** (two germ layers) or **triploblastic** (three germ layers). Triploblastic animals with a body cavity may have a **pseudocoelom** or a true **coelom**.
- **Protostome** and **deuterostome** development often differ in patterns of cleavage, coelom formation, and blastopore fate.

? Describe how body plans provide useful information yet should be interpreted cautiously as evidence of evolutionary relationships.

CONCEPT 32.4

Views of animal phylogeny continue to be shaped by new molecular and morphological data (pp. 676–678)

This phylogenetic tree shows key steps in animal evolution:



? Consider clades Bilateria, Lophotrochozoa, Metazoa, Chordata, Ecdysozoa, Eumetazoa, and Deuterostomia. List the clades to which humans belong in order from the most to the least inclusive clade.

TEST YOUR UNDERSTANDING

LEVEL 1: KNOWLEDGE/COMPREHENSION

- Among the characteristics unique to animals is
 - gastrulation.
 - multicellularity.
 - sexual reproduction.
 - flagellated sperm.
- The distinction between sponges and other animal phyla is based mainly on the absence versus the presence of
 - a body cavity.
 - a complete digestive tract.
 - mesoderm.
 - true tissues.
- Which of the following was probably the *least* important factor in bringing about the Cambrian explosion?
 - the emergence of predator-prey relationships
 - an increase in the concentration of atmospheric oxygen
 - the movement of animals onto land
 - the origin of *Hox* genes

LEVEL 2: APPLICATION/ANALYSIS

- Based on the tree in Figure 32.11, which statement is false?
 - The animal kingdom is monophyletic.
 - Acoelomate flatworms are more closely related to echinoderms than to annelids.
 - Sponges are basal animals.
 - Bilaterians form a clade.

LEVEL 3: SYNTHESIS/EVALUATION

5. EVOLUTION CONNECTION

A professor begins a lecture on animal phylogeny (as shown in Figure 32.11) by saying, “We are all worms.” In this context, what did she mean?

6. SCIENTIFIC INQUIRY

INTERPRET THE DATA Redraw the bilaterian portion of Figure 32.11 for the nine phyla in the table below. Consider these blastopore fates: protostomy (mouth develops from the blastopore), deuterostomy (anus develops from the blastopore), or neither (the blastopore closes and the mouth develops elsewhere). Depending on the blastopore fate of its members, label each branch that leads to a phylum with P, D, N, or a combination of these letters. What is the ancestral blastopore fate? How many times has blastopore fate changed over the course of evolution? Explain.

Blastopore Fate	Phyla
Protostomy (P)	Platyhelminthes, Rotifera, Nematoda; most Mollusca, most Annelida; few Arthropoda
Deuterostomy (D)	Echinodermata, Chordata; most Arthropoda; few Mollusca, few Annelida
Neither (N)	Acoela

Source: A. Hejnol and M. Martindale, The mouth, the anus, and the blastopore—open questions about questionable openings. In *Animal Evolution: Genomes, Fossils and Trees*, eds. D. T. J. Littlewood and M. J. Telford, Oxford University Press, pp. 33–40 (2009).

7. WRITE ABOUT A THEME: INTERACTIONS

Animal life changed greatly during the Cambrian explosion, with some groups expanding in diversity and others declining. Write a short essay (100–150 words) interpreting these events as feedback regulation at the level of the biological community.

8. SYNTHESIZE YOUR KNOWLEDGE



This organism is an animal. What can you infer about its body structure and lifestyle (that might not be obvious from its appearance)? This animal has a deuterostome developmental pattern and a lophophore. To which major clades does this animal belong? Explain your selection, and describe when these clades originated and how they are related to one another.

For selected answers, see Appendix A.

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